Product Specification

Intelligent Ingestion & Attribute Prediction System

eParts Studio Team

Client: eParts Services LLC

Document Version: 1.4

July 29, 2026

Contents

Version History

Version	Date	Change
0.1	Feb 01, 2026	Initial draft based on eParts architectural review.
0.5	Feb 10, 2026	Added detailed functional requirements for Ingestion and ML Service.
1.0	Apr 24, 2026	Baseline specification for development.
1.1	Jul 23, 2026	Integrated ETIM classification/enrichment; corrected OCR (Azure Document Intelligence) and ingestion (channels, Azure Blob storage, quarantine).
1.2	Jul 28, 2026	Pinned the project to ETIM release 10.0 (language EI) for its duration (new constraint C-4); scoped FR-10 accordingly and removed the implied obligation to adopt later ETIM releases.
1.3	Jul 29, 2026	Corrected the placement of ETIM matching. ETIM class/feature/value matching is performed by the ML service *after* attribute matching, not by the Intermediate Structured Layer during normalization. HLR-2, §2.1 and SCEN-1 amended accordingly; FR-9 attributed to the ML service.
1.4	Jul 29, 2026	Closed a trace gap opened by 1.1: the Quality Attribute Scenarios and Validation Requirements sections had not been revised for ETIM. Added QAS-3 (client feature policy applied as configuration, per ADR-019) and VAL-4/VAL-5 (ETIM dictionary load; class review before attribute routing). Also corrected three descriptions left inconsistent by the 1.3 amendment — the Canonical Table glossary entry and §1.2 both still implied ETIM keying during normalization, and §1.2 did not mention ETIM matching at all — and replaced the §2 architecture figure with the current v6.0 diagram. No existing requirement was changed.

Glossary

Term	Definition
PIMS	Product Information Management System. The system where the final product catalog data lives (downstream source of truth). Approved output is written to PIMS keyed by ETIM identifiers.
Ingestion Gateway	The entry point for raw supplier files (CSV and PDF, via SFTP or direct upload; Email and web sources are planned) into the system.

Term	Definition
Confidence Score	A numerical value (0-1) assigned by the ML model indicating the probability that a predicted attribute is correct.
Human-in-the-Loop	A workflow where low-confidence ML predictions are routed to a human operator for manual verification.
ETIM	A standardized technical product-classification model (product groups $\rightarrow$ classes $\rightarrow$ features $\rightarrow$ allowed values / units) used to normalize and enrich supplier attributes. It is reference data layered onto the pipeline, not supplier product data. This project targets ETIM release **10.0**, language **EI**, and does not adopt later releases (C-4).
Canonical Table	A standardized, intermediate database schema used to normalize disparate supplier formats before prediction. It holds the supplier's own product identity and attribute values as evidence. ETIM identifiers are *not* assigned here; the ML service attaches them afterwards into a separate interpretation table (ADR-014).

1 Introduction

1.1 Purpose

The purpose of this system is to automate the ingestion, classification, and attribute extraction of product specifications for eParts Services. Currently, this process is manual, error-prone, and unscalable. This project aims to:

- Standardize disparate supplier data formats (PDF, CSV, Email).
- Utilize Machine Learning to predict product attributes with confidence scoring.
- Implement a "Human-in-the-Loop" workflow to review low-confidence predictions, ensuring high data quality for the PIMS.

1.2 Product Description

The system is a cloud-native data pipeline composed of three primary stages:

1. **Ingestion & Normalization:** A gateway that accepts raw files via SFTP or direct upload (Email and web sources planned), validates them, and converts them into a structured intermediate format holding the supplier's own values. Text-native CSV/PDF are parsed deterministically; datasheet PDFs are OCR'd (Azure AI Document Intelligence) before extraction.
2. **Prediction Service:** An ML-driven engine that matches attributes (e.g., Voltage, Dimensions, Material) and then matches them onto the ETIM classification standard, assigning a confidence score at each step.
3. **Review & Writeback:** A routing layer that auto-accepts high-confidence data into PIMS and flags low-confidence data for a React-based Human Review UI.

1.3 Stakeholders & Personas

[PS-1] Persona: The Ops Reviewer (Internal)

The Ops Reviewer is a non-technical domain expert at eParts. They are responsible for ensuring catalog accuracy but are currently overwhelmed by manual data entry.

- **Goal:** Review and approve items quickly without needing to manipulate raw SQL or JSON.
- **Pain Point:** Spending hours manually copying data from PDFs to Excel.
- **Need:** A clean UI that highlights exactly *what* needs review (low confidence items) and links back to the source document.

[PS-2] Persona: The Supplier (External)

The Supplier is a manufacturer providing product catalogs. They utilize various legacy formats and are resistant to changing their own internal systems.

- **Goal:** Submit product data with minimal friction.

- **Constraint:** Will not adhere to a strict API schema; sends "messy" CSVs or PDFs via Email/SFTP.

2 System Architecture

The system follows a microservices-based pipeline architecture designed for breadth-first delivery.

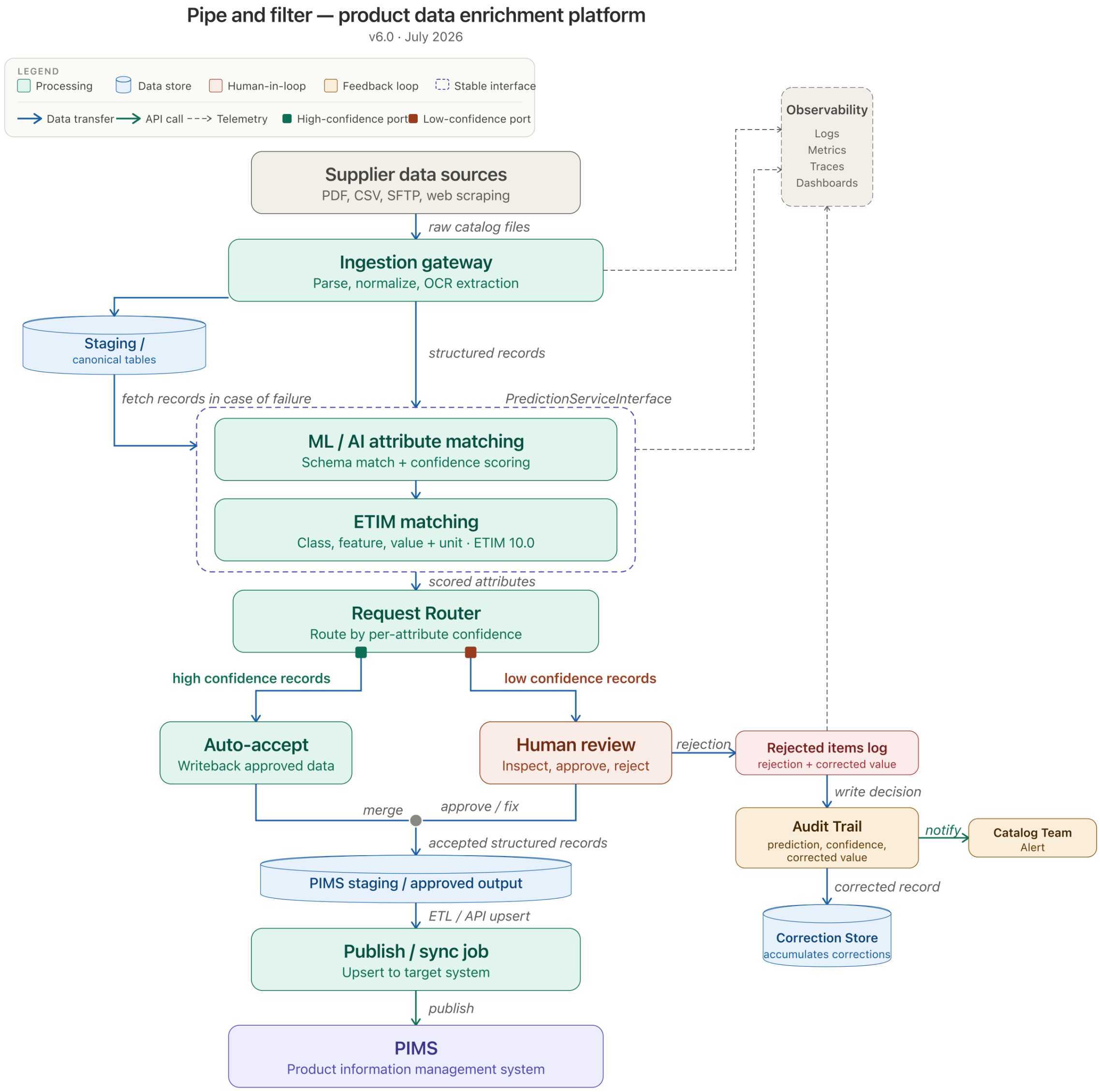

Figure 1: eParts reference architecture, v6.0. Solid outlines are running code; dashed outlines are designed and not yet built.

2.1 Component Descriptions

- **Ingestion Gateway:** Handles "messy" inputs. Identifies submission source, creates ingestion records, and archives raw files to Azure Blob Storage for auditability. Datasheet PDFs are OCR'd via Azure AI Document Intelligence; text-native CSV/PDF are parsed deterministically.
- **Intermediate Structured Layer:** Converts raw inputs into standard canonical

tables holding the supplier's own values as evidence (product identity plus raw attributes). It performs mechanical cleanup and schema-based transformation only; it does *not* assign ETIM identifiers.

- **Attribute Prediction Service (ML):** Owns all matching, in two phases. **Attribute matching** maps a raw supplier label and value onto an attribute identity, value, and confidence. **ETIM matching** then runs against the ETIM reference dictionary: class match, feature match, value and unit normalization, ETIM validation, and client-policy validation, attaching a confidence score to each ETIM assignment. Outputs the original value as evidence alongside the ETIM identifiers.
- **Confidence Routing:** A logic layer that splits traffic. High confidence → Auto-Accept. Low confidence → Human Review Queue.
- **Engineering Ops / Observability:** Centralized logging (datadog) and metrics to track model drift and ingestion success rates.

3 Operational Environment

3.1 Cloud Infrastructure

- **Hosting:** The system shall be hosted on public cloud (Azure) using containerized services (Kubernetes/Docker).
- **Database:**
  - **Transactional:** PostgreSQL/MSSQL for user accounts, workflow state, and intermediate tables.
  - **Data Warehouse:** (Future) for historical model training data.
- **Document Extraction:** Azure AI Document Intelligence (layout OCR) with Azure OpenAI for datasheet attribute extraction.
- **Object Storage:** Azure Blob Storage archives raw supplier files for traceability, accessed through an S3-compatible interface (MinIO in local/dev).
- **Authentication:** Auth0 must be used for all internal user sign-in and role-based access control (RBAC).

3.2 System Constraints

ID	Constraint
C-1	**Cost-Effective Design:** Avoid architectures where cost grows linearly per tenant; prefer shared resources where possible.
C-2	**Privacy Compliance:** The system must support data deletion requests (GDPR/CCPA) for specific supplier submissions.
C-3	**Breadth-First Delivery:** The initial release must demonstrate a full end-to-end flow for a single supplier type before optimizing model depth.
C-4	**ETIM Release Pinned:** The system shall target ETIM release 10.0 (language EI) for the duration of this project. Adopting later ETIM releases, and migrating already-classified products between releases, are out of scope.

4 Requirements

4.1 High Level Requirements (HLR)

ID	High Level Requirement
HLR-1	The system shall ingest product data from diverse supplier sources (SFTP and direct upload today; Email and web sources planned) in CSV and PDF formats.
HLR-2	The system shall normalize ingested data into a standardized intermediate structure through mechanical cleanup only, preserving original supplier values as evidence. ETIM class, feature, and value assignment is performed downstream by the ML service (HLR-6, FR-9), not during normalization.
HLR-3	The system shall predict product attributes and assign confidence scores using Machine Learning.
HLR-4	The system shall provide a user interface for human review of low-confidence predictions.
HLR-5	The system shall write approved data back to the PIMS (Product Information Management System).
HLR-6	The system shall classify products against the ETIM standard and enrich supplier attributes with ETIM identifiers (class, feature, value, unit), keeping the original values as evidence.

4.2 Functional Requirements (FR)

ID	Requirement (EARS Format)
FR-1	Upon receipt of a file via SFTP or direct upload (Email and web ingestion are planned), the Ingestion Gateway shall create an ingestion record containing supplier ID, timestamp, and source channel.
FR-2	The Ingestion Gateway shall validate basic file integrity (type, size, virus scan) before processing; records that fail validation shall be routed to a quarantine store for triage rather than silently dropped.
FR-3	The Attribute Prediction Service shall generate a confidence score (0.0 to 1.0) for every predicted attribute.
FR-4	If a prediction's confidence score is below the configured threshold, the system shall route the item to the Human Review Queue.
FR-5	The Human Review UI shall display the predicted value alongside the original source snippet (text/image) for verification.
FR-6	The system shall log every human review action (accept/edit/reject) to an immutable audit trail.
FR-7	The system shall allow authorized Ops Leads to adjust the confidence threshold for auto-acceptance.

ID	Requirement (EARS Format)
FR-8	Upon final approval (auto or human), the system shall write the attributes to PIMS via API.
FR-9	After attribute matching, the Attribute Prediction Service (ML) shall match attributes to ETIM classes, features, and controlled values/units, attaching a confidence score to each ETIM assignment and preserving the original supplier value.
FR-10	The system shall load and maintain the ETIM reference dictionary (product groups, classes, features, values, units, and class–feature–value mappings) as reference data for the pinned ETIM release identified in C-4.

4.3 Derived Requirements (DR)

ID	Priority	Requirement	Trace
DR-1	Must	The system shall archive the original raw file in Azure Blob Storage as "evidence" for audit purposes even after processing.	HLR-1
DR-2	Should	The system shall support manual triggering of model retraining using corrected data from the Review Queue.	HLR-3
DR-3	Must	The Writeback service must be idempotent; retrying a write to PIMS must not create duplicate records.	HLR-5
DR-4	Must	Approved data written to PIMS shall be keyed by ETIM identifiers (release, class, feature); the writeback idempotency key shall include these identifiers.	HLR-6

5 Quality Attribute Scenarios

QAS-1: Modifiability (Extensibility)

Attribute	Scenario
ID	QAS-1
Attribute	Modifiability
Source	Developer
Stimulus	A new supplier format (e.g., a new JSON schema) needs to be added to the pipeline.
Response	The developer adds a new mapping configuration without rewriting the core pipeline code.
Measure	A new supplier format can be integrated and deployed within 4 hours of engineering effort.

QAS-2: Usability (Ops Efficiency)

Attribute	Scenario
ID	QAS-2
Attribute	Usability
Source	Ops Reviewer
Stimulus	The queue contains 100 low-confidence items requiring review.
Response	The UI presents items with "diff" views and hotkeys for acceptance.
Measure	The reviewer can process simple accept/reject decisions at a rate of 10 items per minute.

QAS-3: Modifiability (Client Feature Policy)

Attribute	Scenario
ID	QAS-3
Attribute	Modifiability
Source	Client / Ops Lead
Stimulus	The client changes which ETIM features are mandatory for a given product class. ETIM itself carries no required-field flag, so this policy is ours to hold.
Response	The change is applied as per-class configuration; no matching, routing or validation code is modified (ADR-019).
Measure	The revised policy takes effect for the next ingested batch without a code deployment.

6 Operational Scenarios

[SCEN-1] End-to-End Ingestion (Happy Path)

- **Objective:** A supplier submits a catalog, and it is processed into PIMS without human intervention.
- **Actors:** Supplier, System

Source	Step #	Req.	Action
Supplier	1	-	Uploads a CSV catalog to the designated SFTP folder.
System	2	FR-1	Detects file, creates ingestion record, and archives raw file.
System	3	HLR-2	Cleans and normalizes CSV columns into the Canonical Table, retaining the supplier's original values as evidence. No ETIM assignment happens here.
System	4	FR-3, FR-9	ML Service matches attributes, then matches them to ETIM class, features, and values. All confidence scores are > 0.90.
System	5	FR-8	Auto-Accept logic triggers. Data is written to PIMS.

[SCEN-2] Low Confidence & Review (Human-in-the-Loop)

- **Objective:** A messy PDF is ingested, low confidence is flagged, and a human corrects it.
- **Actors:** System, Ops Reviewer

Source	Step #	Req.	Action
System	1	FR-1	Ingests a datasheet PDF; routed to Azure AI Document Intelligence (layout OCR) with LLM extraction. Raw OCR output is noisy.
System	2	FR-3	ML Service predicts "Voltage: 12V" with confidence 0.45.
System	3	FR-4	Confidence $<$ Threshold (0.80). item routed to Review Queue.
Ops User	4	FR-5	Opens Review UI. Sees "12V" predicted but PDF says "24V".
Ops User	5	FR-6	Corrects value to "24V" and clicks Approve. Action logged.
System	6	FR-8	Validated data "24V" is written to PIMS.

7 Design Constraints & Validation

7.1 Design Constraints

ID	Constraint	Source
DC-1	**Tech Stack:** Backend must be Python based (for ML library compatibility).	Team Charter
DC-2	**Auth:** Must use Auth0 for identity management.	Client Req
DC-3	**Audit:** Raw files must be preserved in Azure Blob Storage for re-processing and traceability.	Regulatory

7.2 Validation Requirements

Test ID	Description	Steps	Expected Result
VAL-1	**Ingestion Trigger**	Upload file to SFTP.	Ingestion record appears in DB within 30 seconds.
VAL-2	**Routing Logic**	Mock ML response with Conf=0.2.	Item appears in Human Review Queue.
VAL-3	**PIMS Integration**	Approve item in UI.	API call to PIMS returns 200 OK.
VAL-4	**ETIM Dictionary Load** (FR-10)	Load the ETIM 10.0 (EI) release files, then run the same load again.	All 159 groups, 5,640 classes, 17,377 features and 201,284 class-feature-values are present, and the second run is a no-op.
VAL-5	**Class Review Before Attribute Routing** (FR-9)	Supply a product whose ETIM class assignment falls below the class-confidence threshold.	The item is routed to class review and no attribute-level routing is performed for it.

VAL-4 is covered by `tests/unit/test_etim_loader.py` — 10 tests, all passing — which exercise the load, the row counts and the idempotent re-import against an in-memory database. `tests/integration/test_etim_real_files.py` performs the same load against the real ETIM 10.0 archive; it skips automatically unless that archive is present locally, and the archive is not committed, so it does not run on a clean checkout. Neither runs in CI. VAL-5 is specified but not yet executable: the ETIM matching stages are designed and not built, so the test exists as a specification only. This is recorded rather than deferred silently.